

Transfer Learning zur Klassifikation von Galaxien

Deep Learning - Station 2 – Transfer Learning

Alexander Nungesser



Agenda

Datensatz

Retrospektive Station 1

ResNet

Inception V3

Zoobot

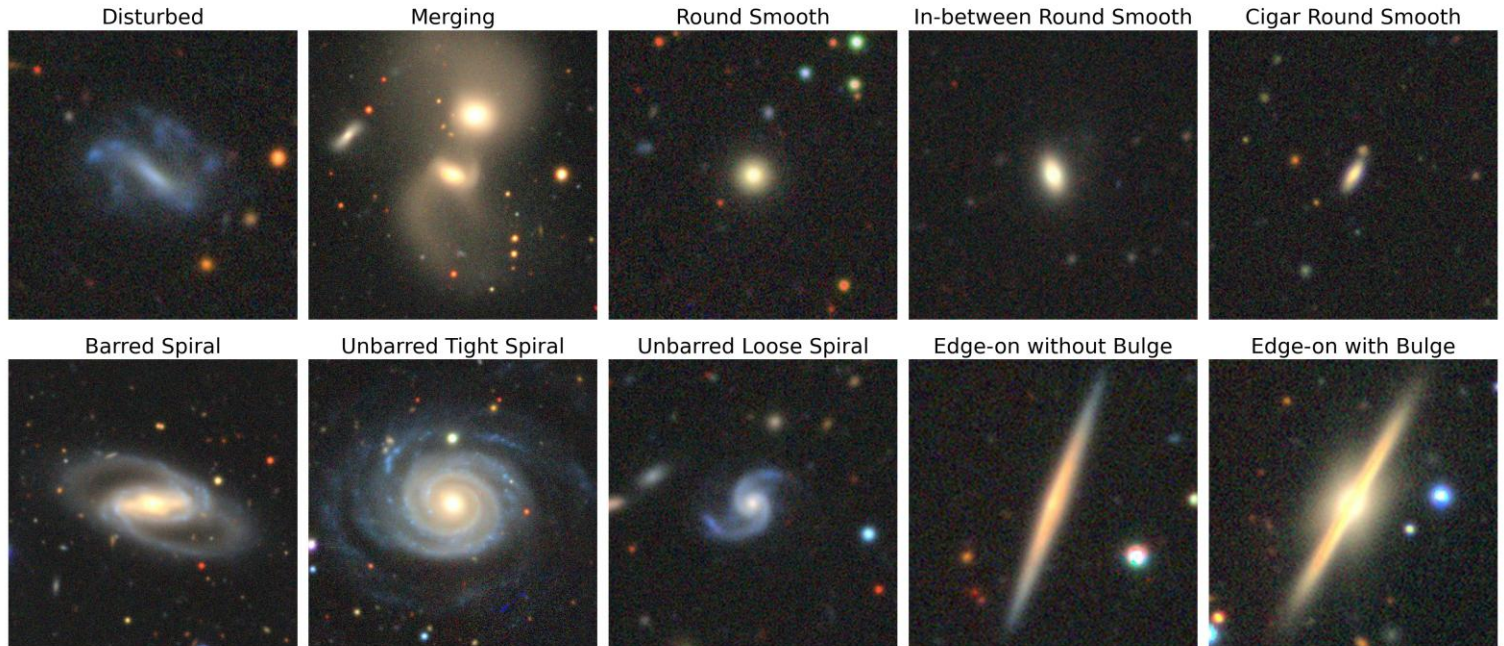
Evaluation

Zusammenfassung & Ausblick

Datensatz

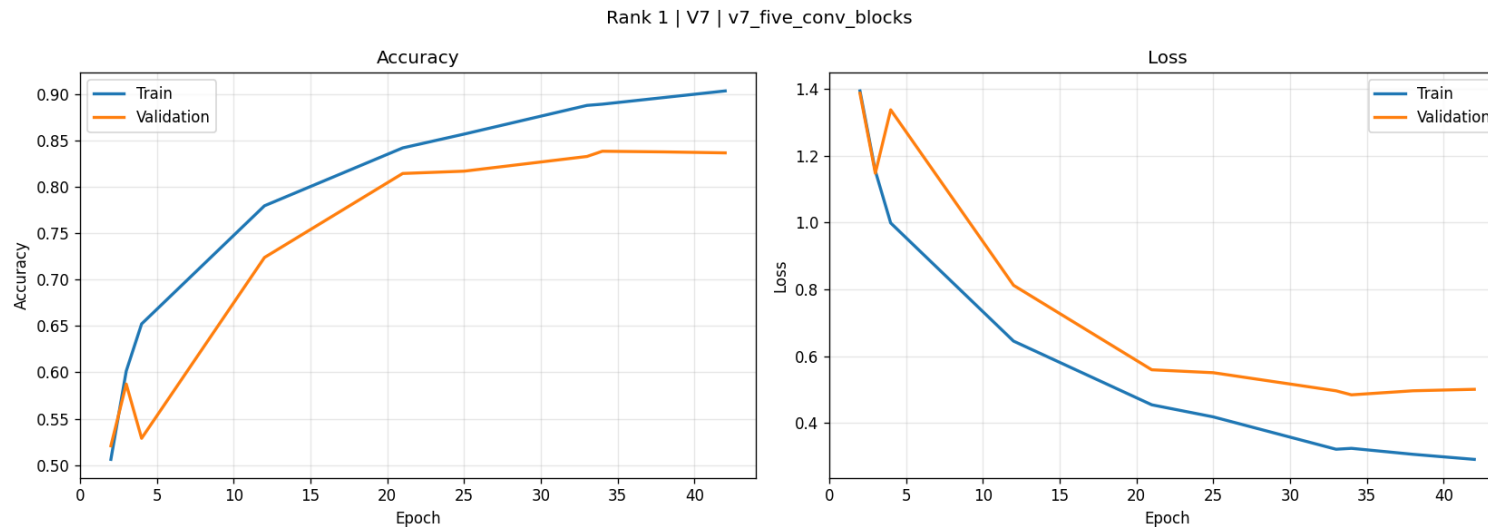
- Umfang:
 - 17.736 Bilder
- Format:
 - 256x256 Pixel
- Kategorisierung:
 - 10 Klassen
- Quellen:
 - *Bilder*: DESI Legacy Imaging Surveys
 - *Labels*: Galaxy Zoo

Example images of each class from Galaxy10 DECals

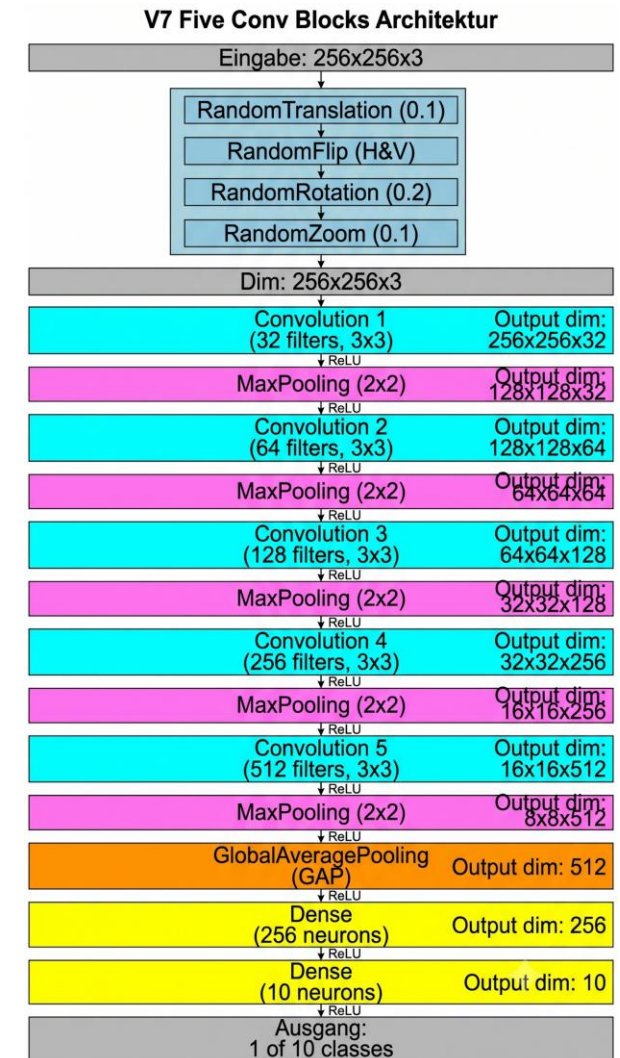


Galaxy10 DECals: Henry Leung/Jo Bovy 2021, Data: DECals/Galaxy Zoo

Retrospektive Station 1

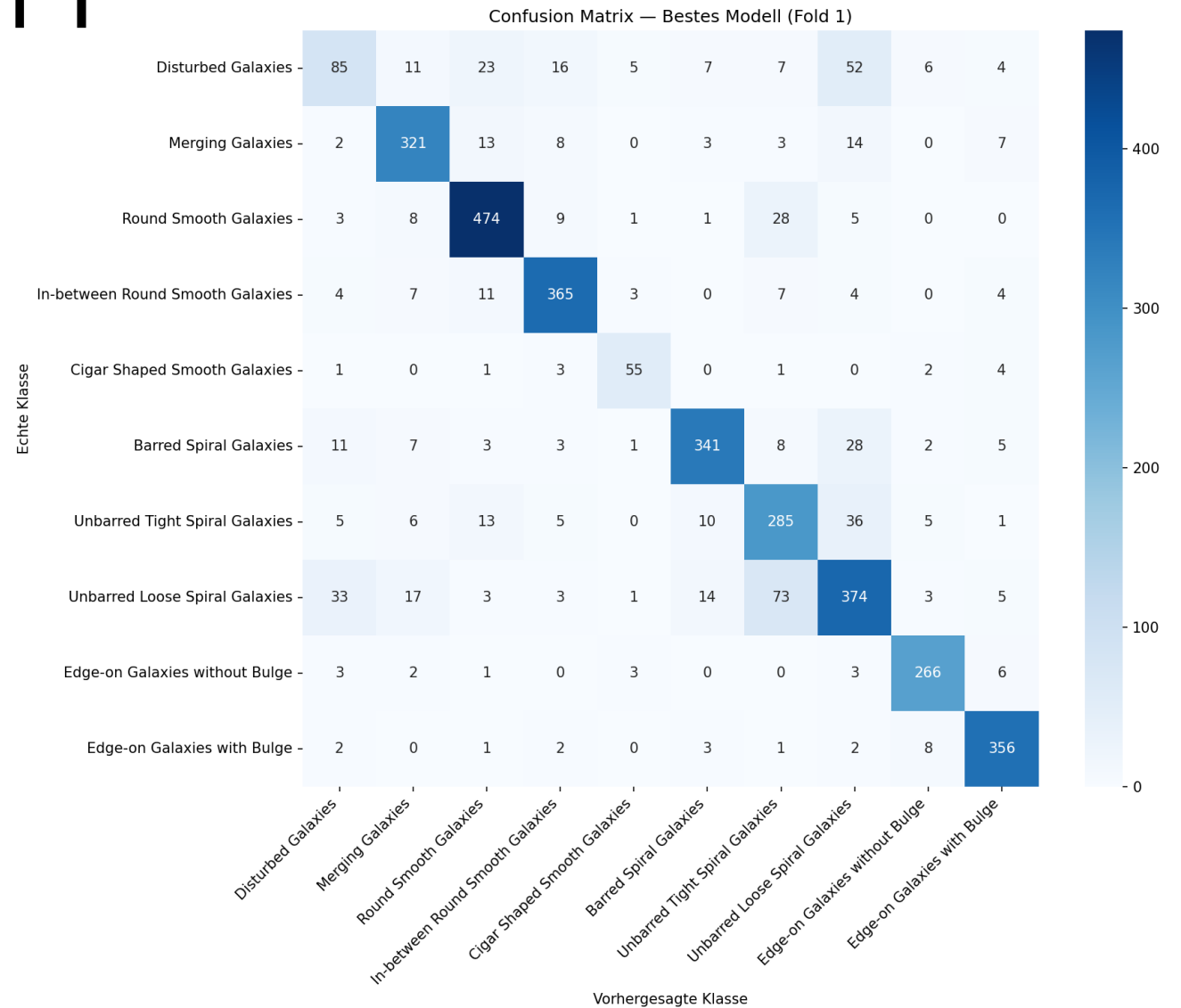


- 82,36% Test-Accuracy
- 2,49 Mio. Parameter
- Overfitting



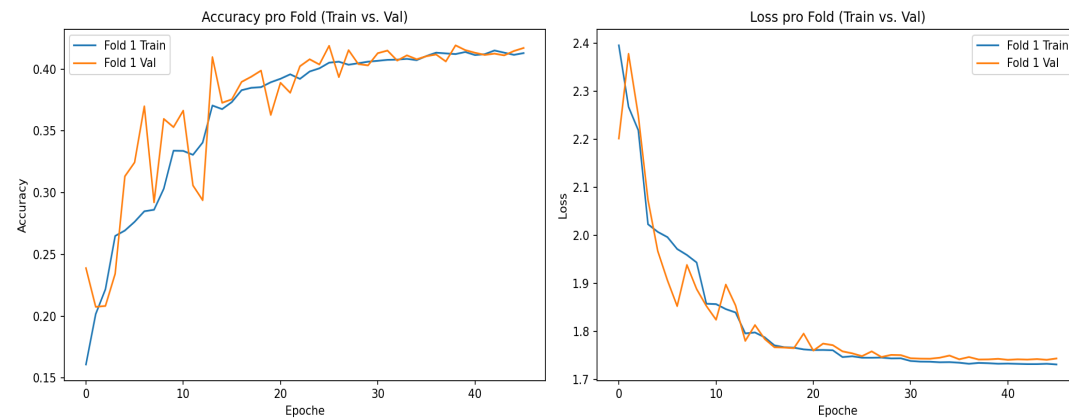
Retrospektive Station 1

- Meist korrekte Klassifizierung
- „Disturbed Galaxies“ keine einheitliche Form
- Verwechslung bei Spiralen
 - Tight vs loose spiral

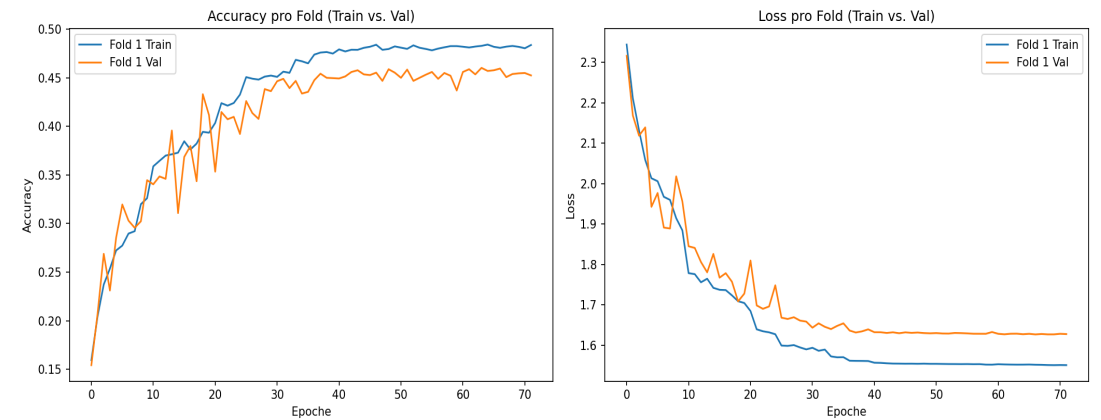


ResNet

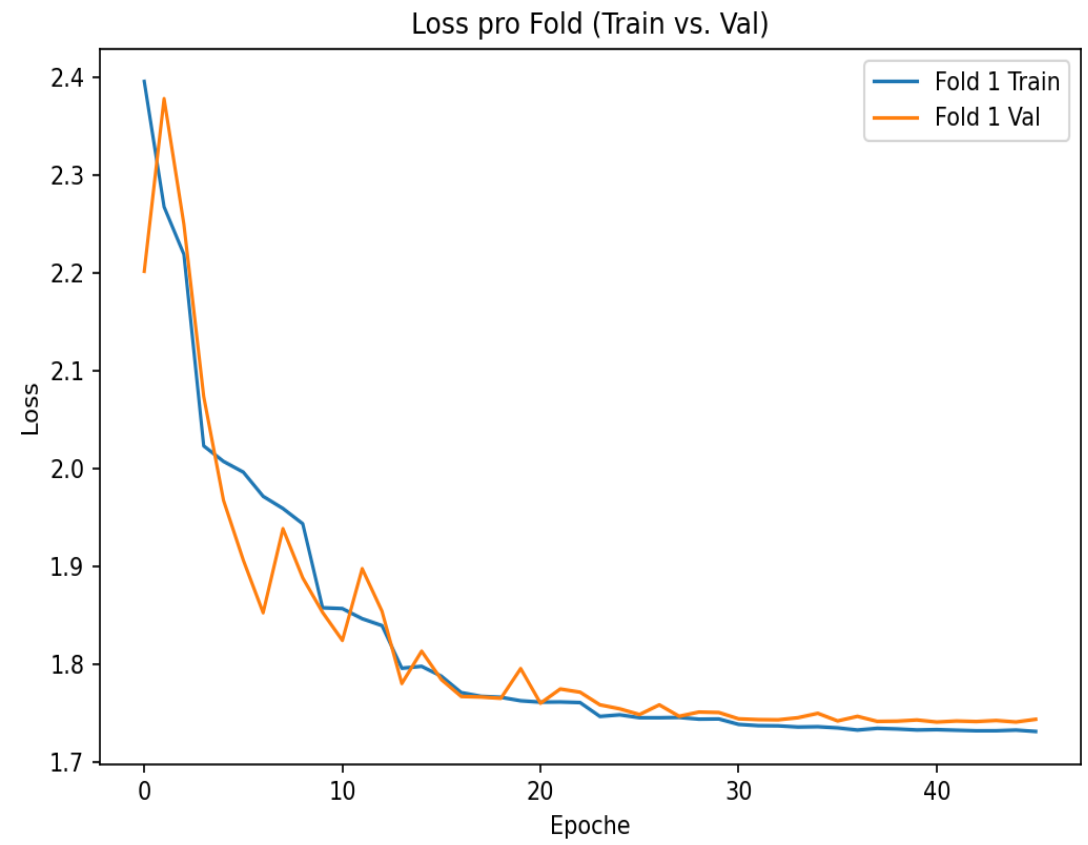
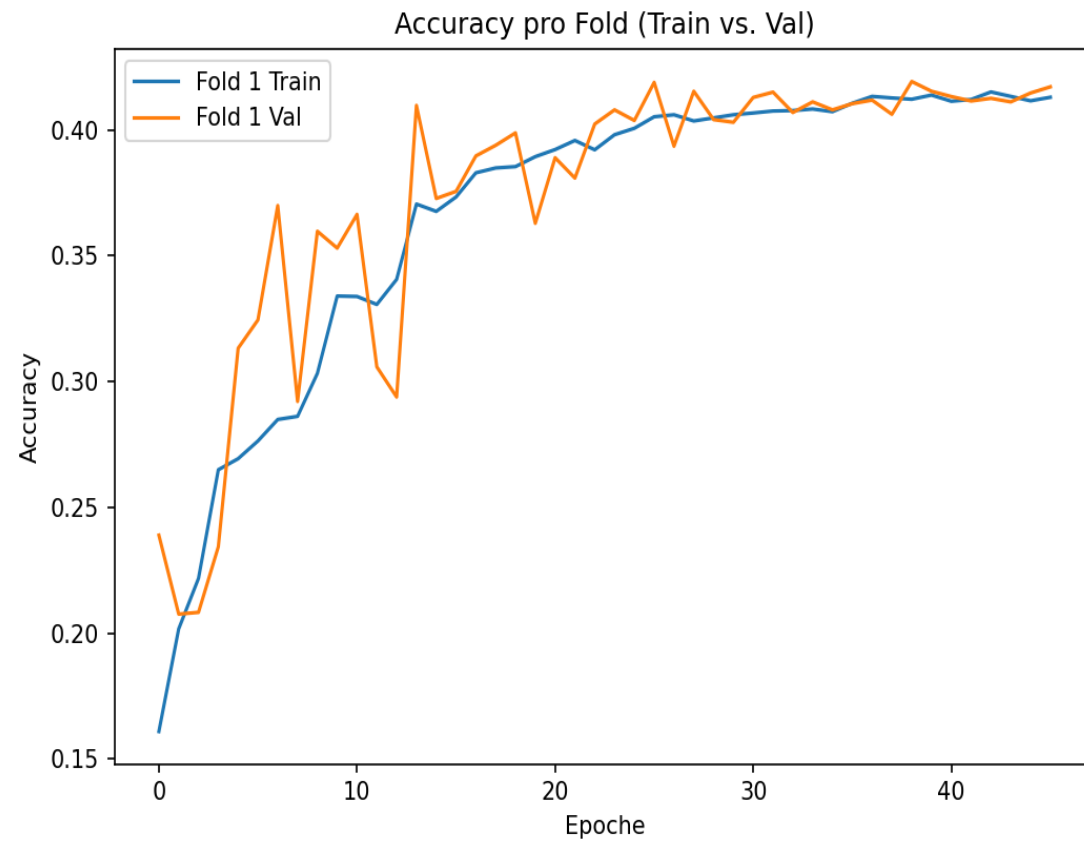
ResNet-50



ResNet-152

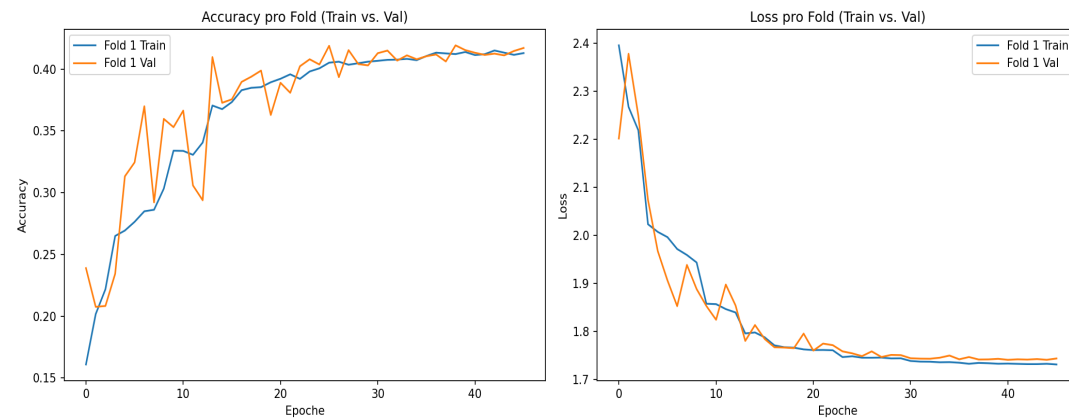


ResNet-50

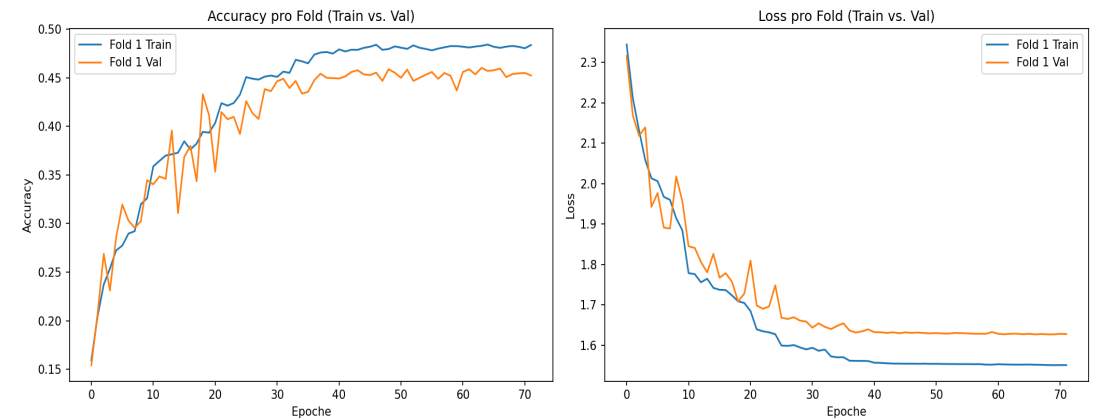


ResNet

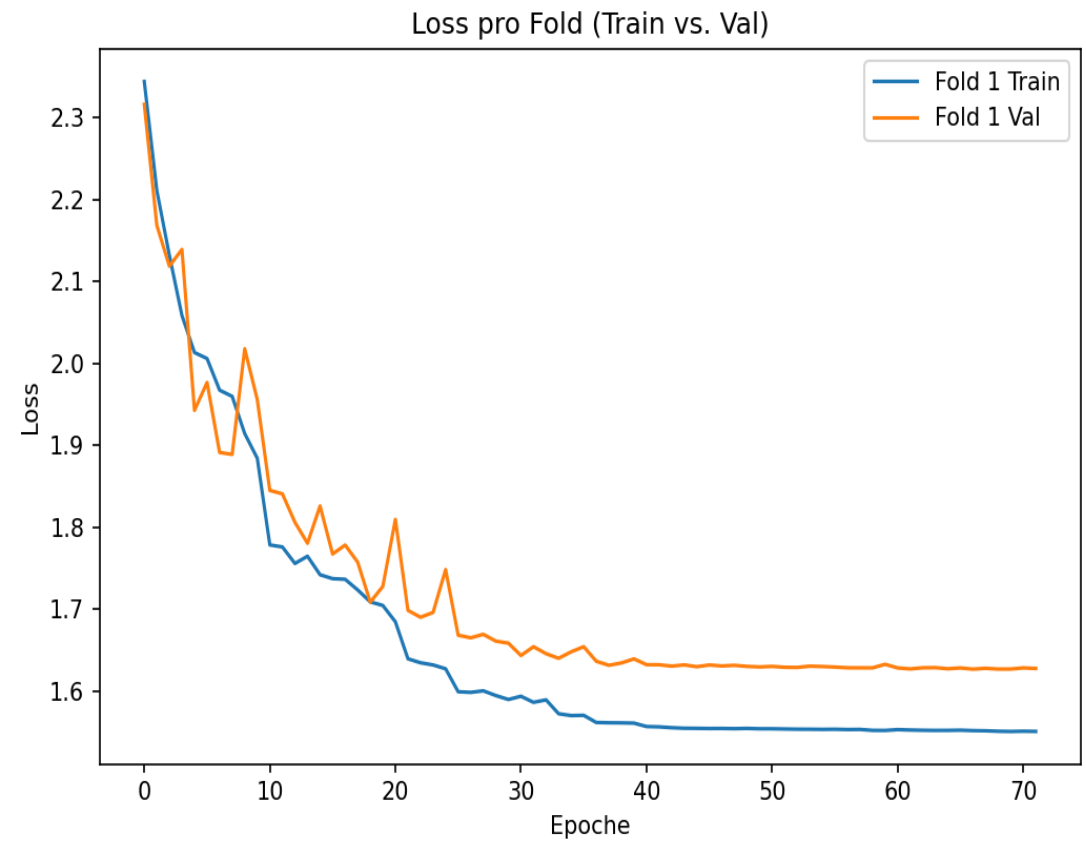
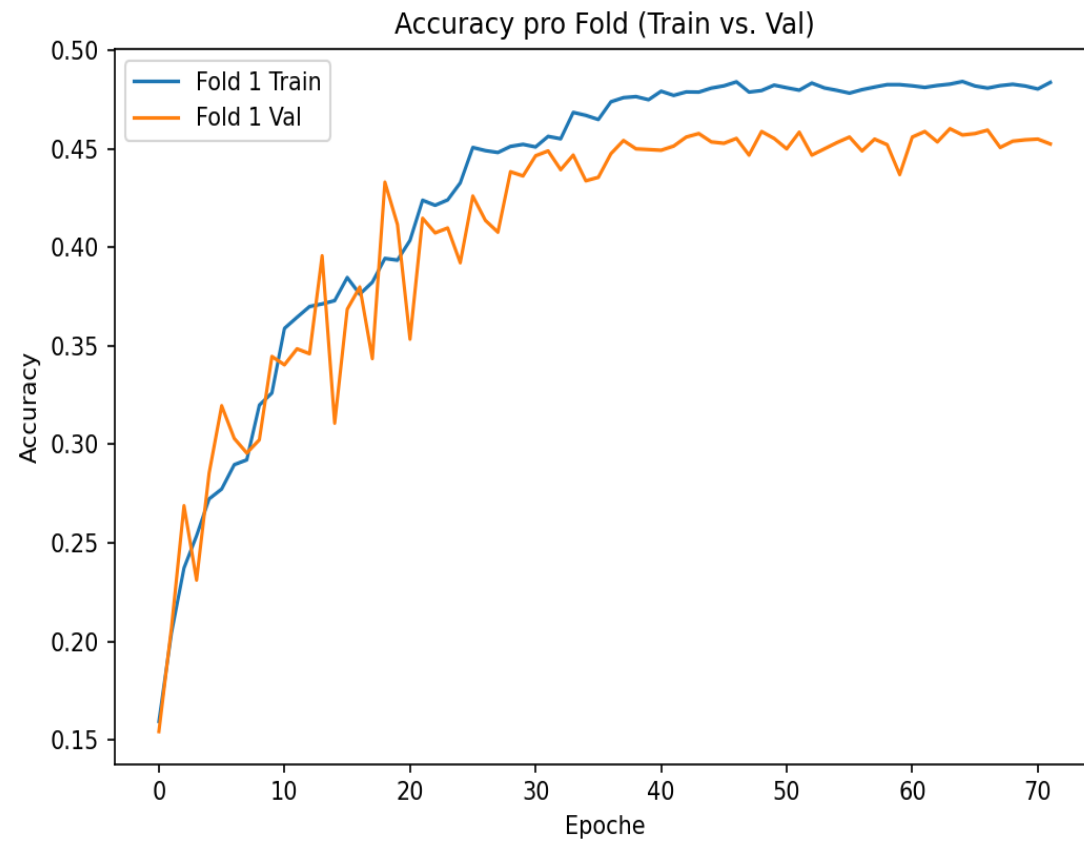
ResNet-50



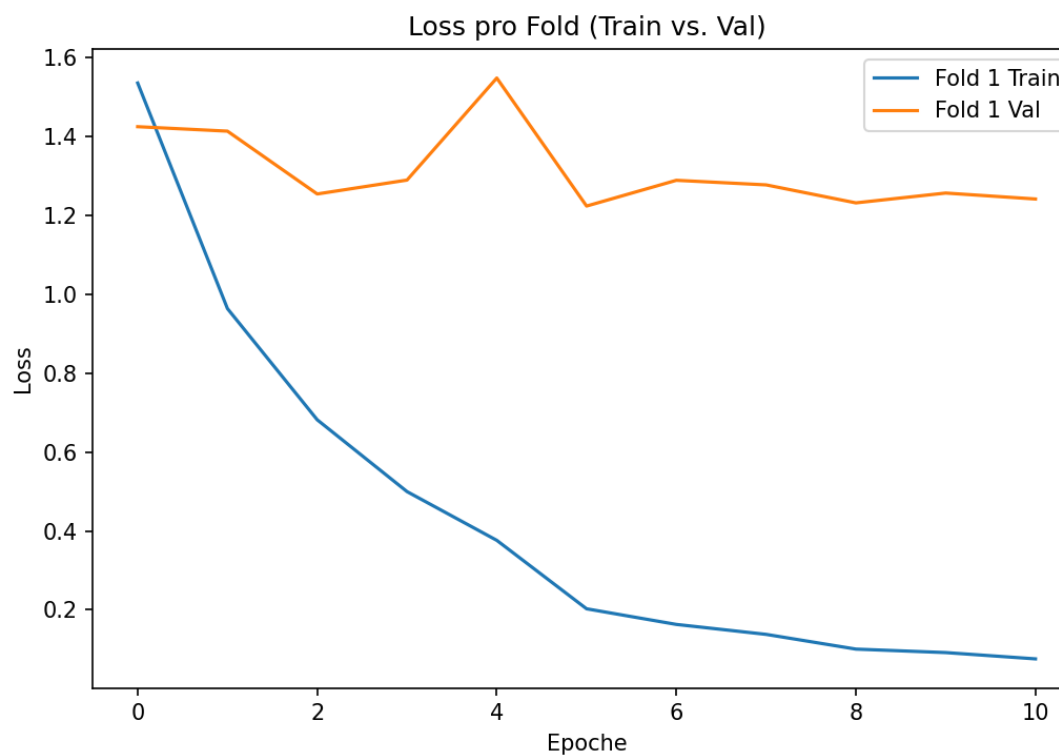
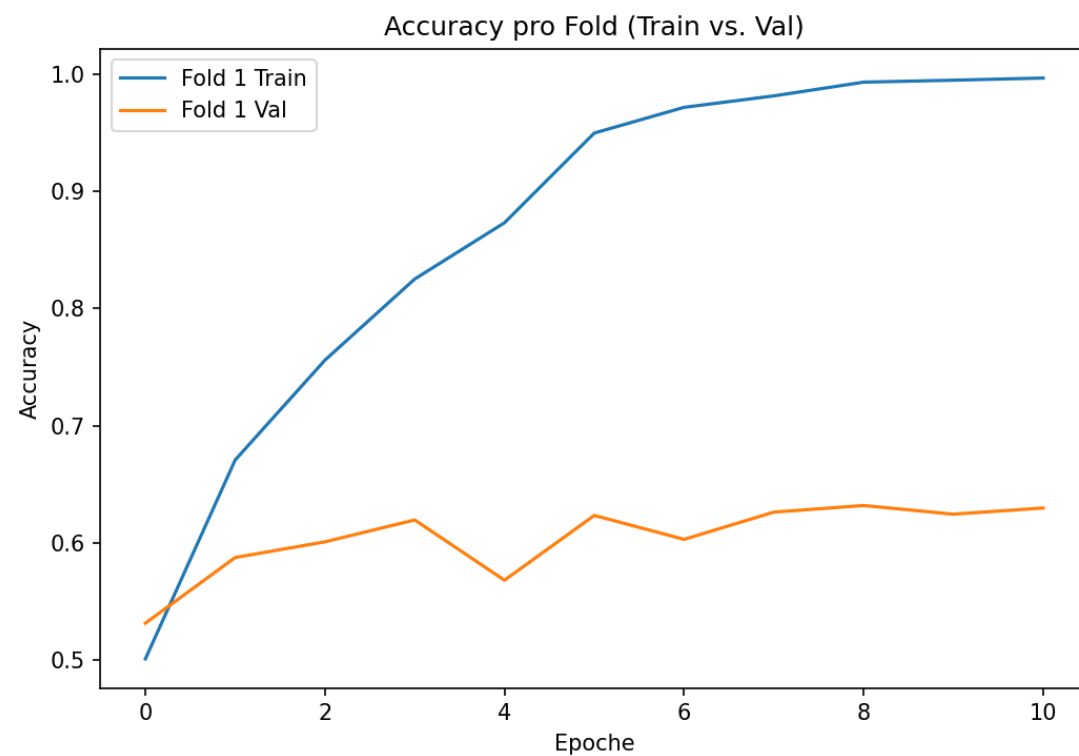
ResNet-152



■ ResNet-152



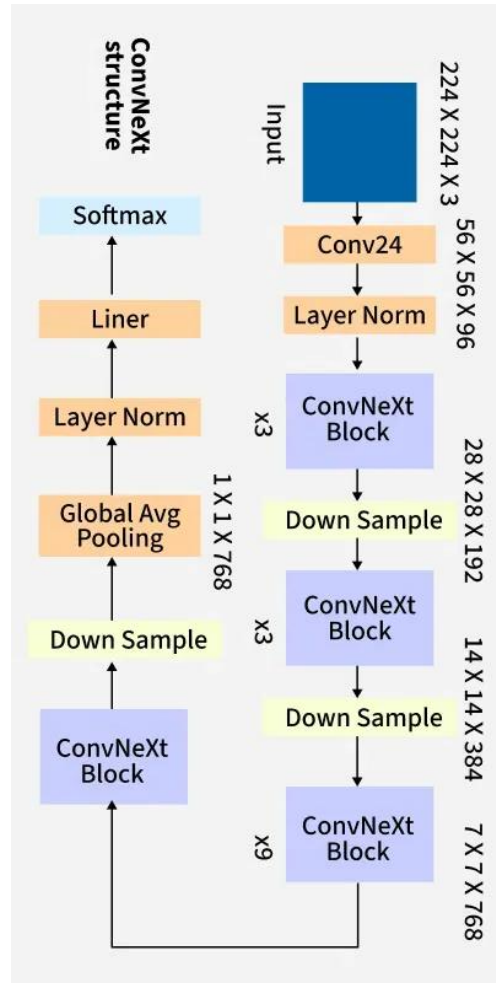
Inception V3



I Zoobot

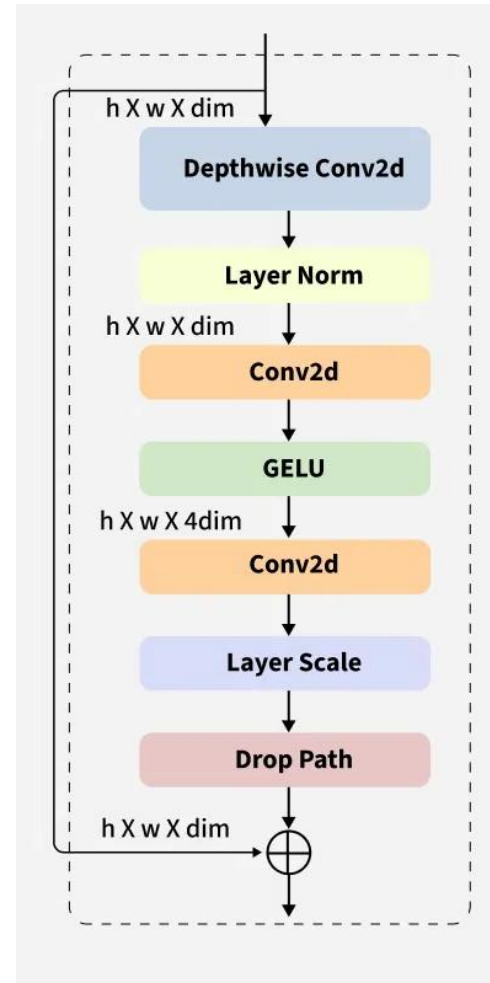
- Vortrainiert auf Galaxien-Bilder
 - Gleiche Domäne
- Konzepte von Vision Transformern (LayerNorm, GELU, 7x7 Kernels)
 - Ohne Self-Attention
- ConvNeXt-Nano
- 15,6 Mio. Parameter

Zoobot



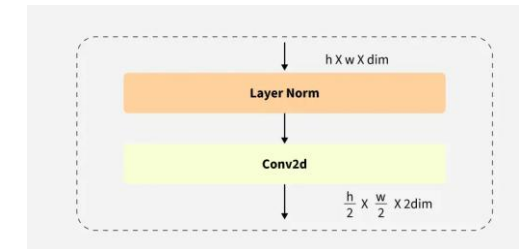
ConvNeXt

09.06.2026



ConvNeXt Block

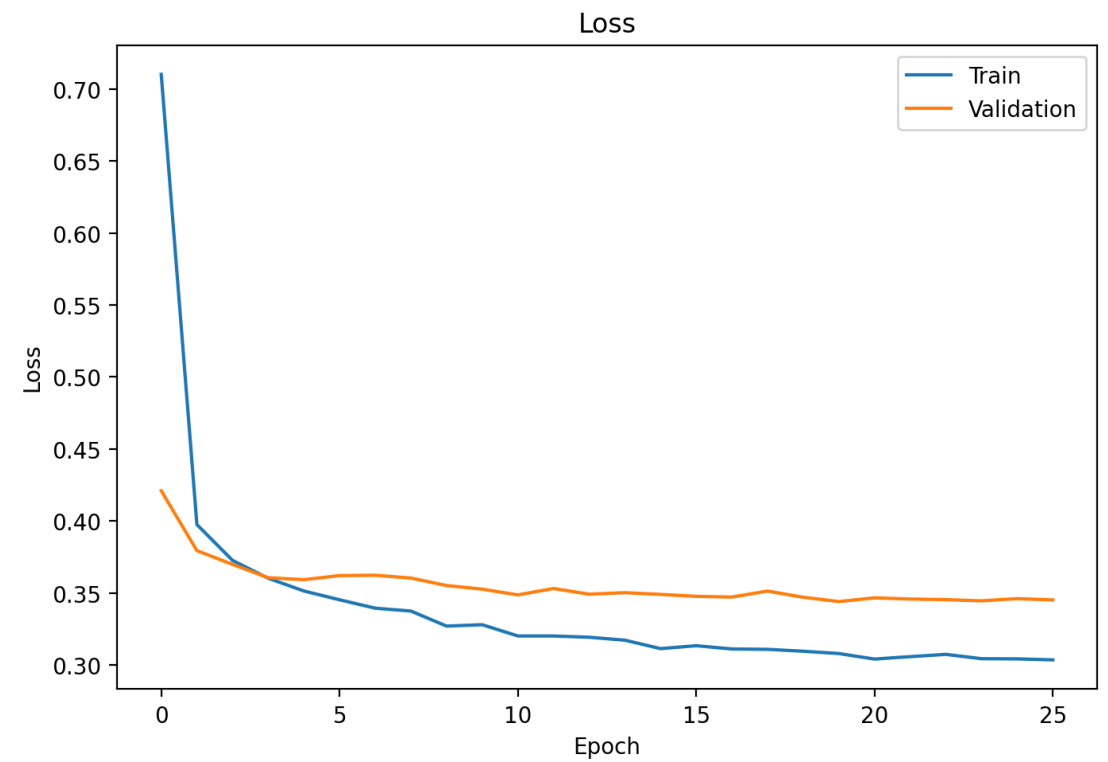
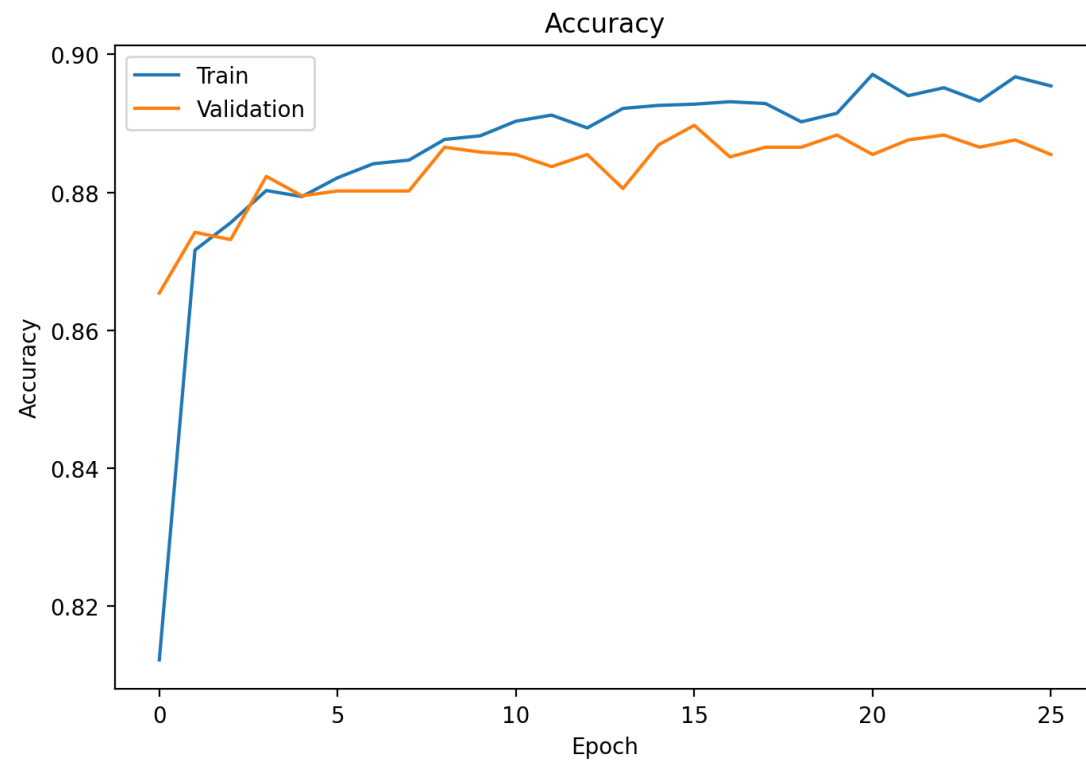
Deep Learning - Station 2 - Transfer Learning



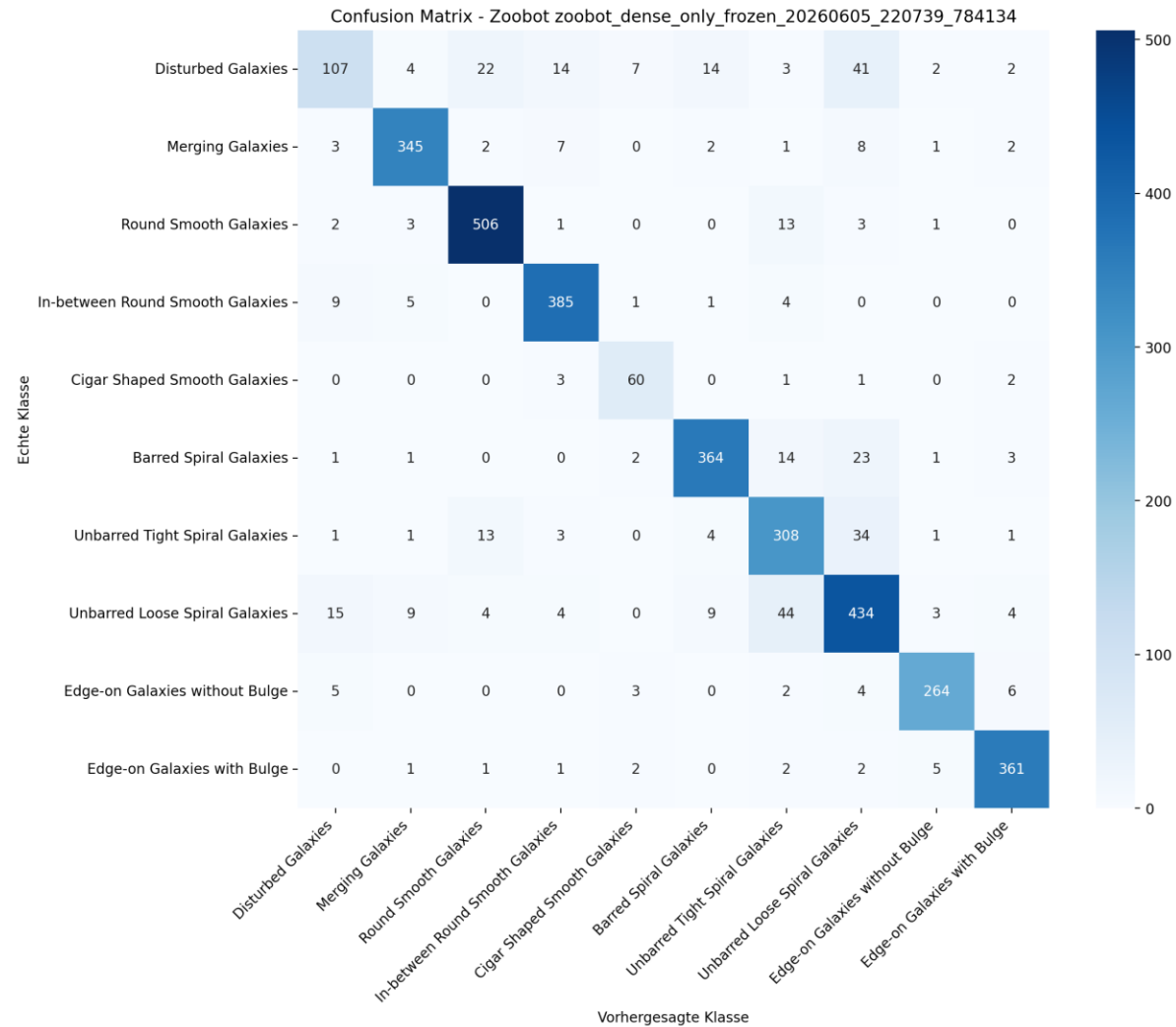
Down Sample

12

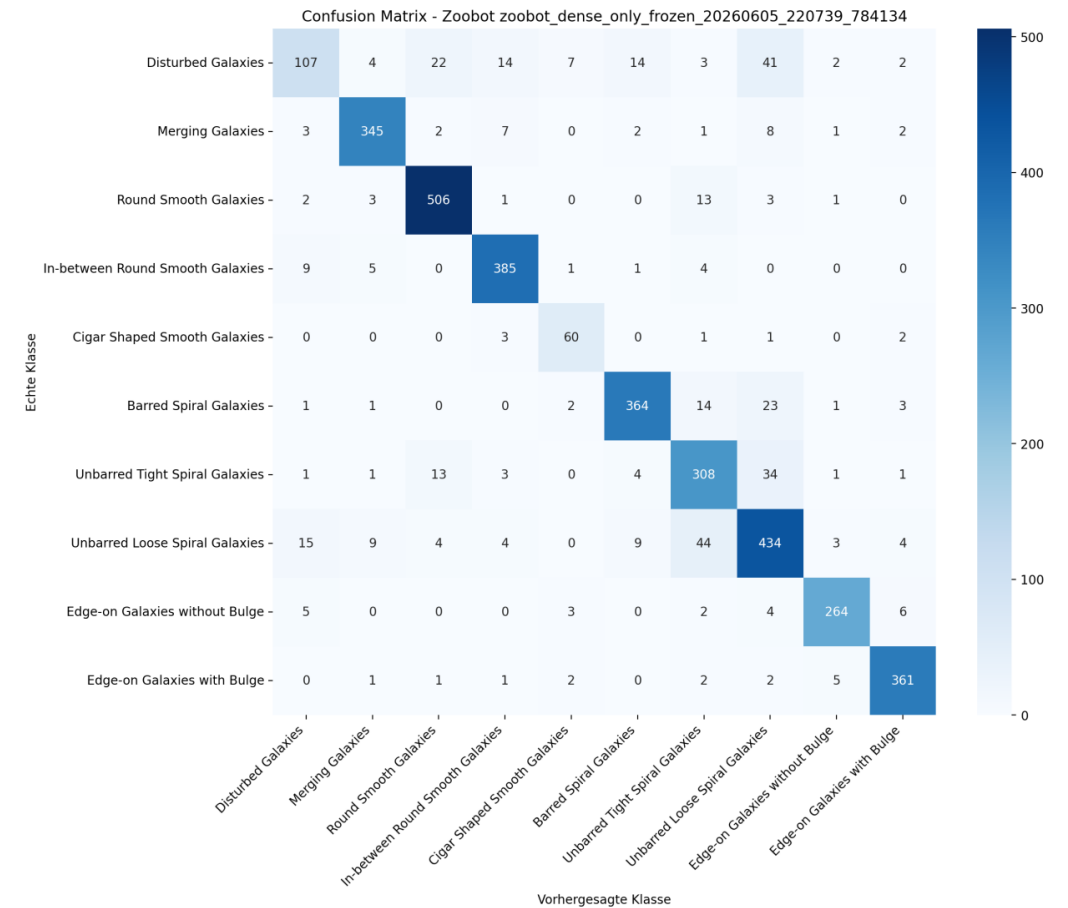
Zoobot



Zoobot



Evaluation

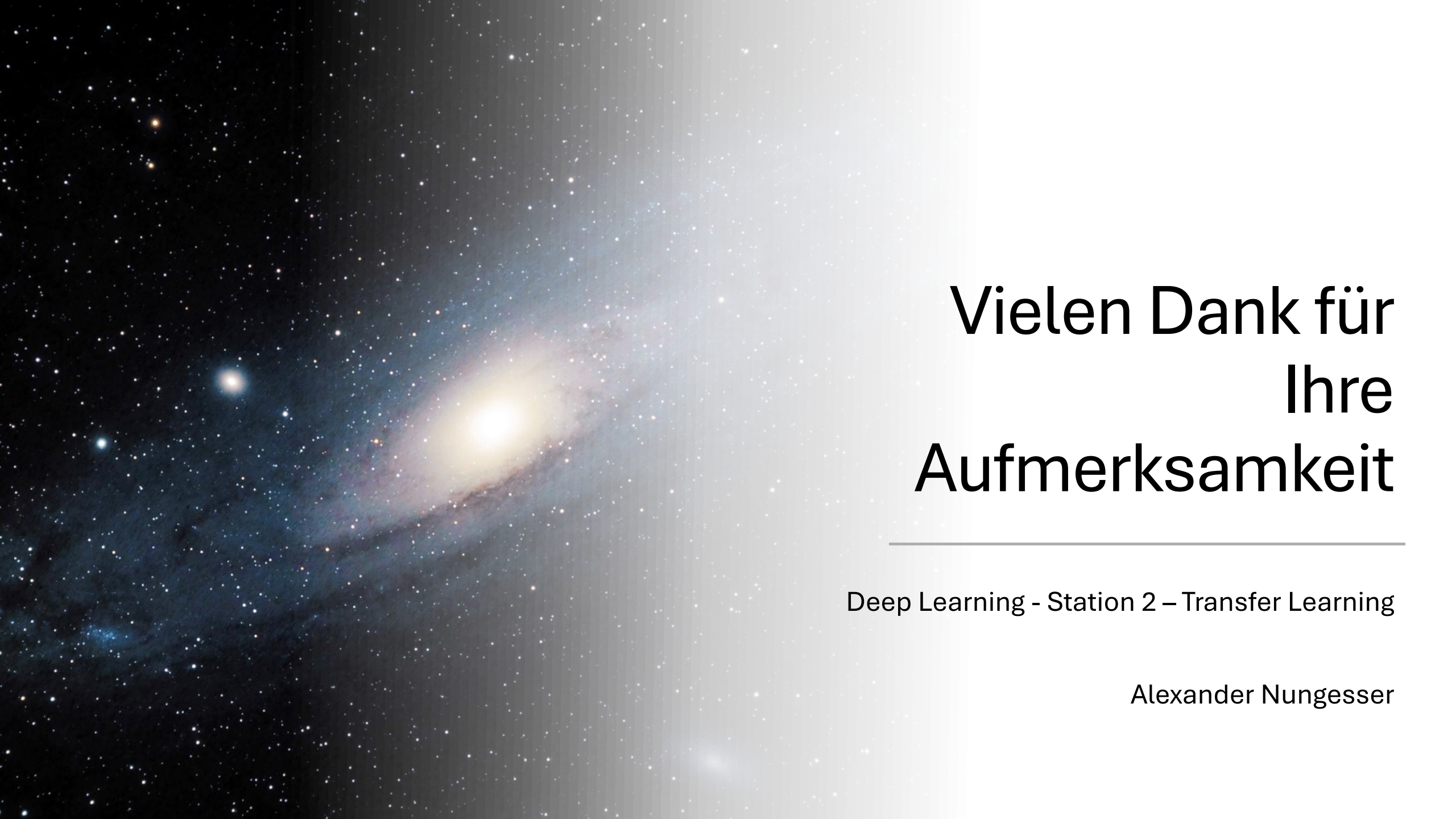


Evaluation

	Station 1	ResNet-50	ResNet-152	Inception V3	Zoobot (ConvNeXt- Nano)	Zoobot (ConvNeXt- Nano) finetuned
Parameter total [Mio.]	2,49	24,89	59,68	22,54	15,11	15,11
Parameter trainierbar [Mio.]	2,49	1,31	1,31	0,73	0,16	15,11
Epochen	41	41	67	6	25	9
Train. Acc. [%]	90,33	41,30	48,37	99,67	89,54	91,52
Val. Acc. [%]	83,65	41,72	45,24	62,97	88,55	89,11
Test Acc. [%]	82,36	38,64	43,55	61,22	88,33	89,12

I Zusammenfassung & Ausblick

- Zoobot liefert bestes Ergebnis
 - Test Acc. +6% vgl. Station 1
- Domänenspezifisches Model erzielt bessere Ergebnisse
 - Test Acc. bis zu +50% vgl. Generische Modelle
- Finetuning
 - Nicht viel Verbesserung
- Zoobot als Teacher für Modelle kleiner als in Station 1



Vielen Dank für Ihre Aufmerksamkeit

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